



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

PROG103: PRINCIPLES OF SOFTWARE ENGINEERING

Title : GROUP PROJECT

Due date : WEEK 12

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Class : DIT1203F

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Group : Group 2

Academic Honesty Policy Statement

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INTRODUCTION

Agriculture is one of the most important sectors in Sierra Leone and provides livelihoods for many rural communities. However, small-scale farmers often face difficulties accessing profitable markets. Due to transportation costs, limited financial resources, and lack of market information, many farmers are unable to bring their produce to major cities where demand and prices are higher, example people in (Bo)

FarmYad is a Digital Public Good System designed to bridge the gap between farmers and buyers by providing an online platform where farmers can advertise their products and buyers can purchase directly from them. The system will reduce middlemen, increase farmers' income, and improve food distribution across the country.

PROBLEM DEFINATION

Problem

Farmers are in the province, producing huge amount of farm produce at the end of every farm season, but they are unable to transport their product to the city for sale due to lack of knowledge in running the farming business and or financial challenges that has limit them to reach certain locations with these products. Many small-scale farmers in rural provinces face several of these challenges:

- ❖ Lack of transportation to major cities
- ❖ Limited access to larger markets
- ❖ Dependence on liaison who pay very low prices
- ❖ Difficulty finding buyers before harvesting

At the same time, buyers in cities struggle to:

- ❖ Find fresh produce directly from farmers
- ❖ Know product availability and pricing
- ❖ Source agricultural products reliably

PROPOSED SOLUTION

FarmYad is a Digital Public Good platform that connects rural farmers directly with city buyers through a web and mobile system.

The platform allows:

- ❖ Farmers to list available produce
- ❖ Buyers to search and purchase products
- ❖ Communication between buyers and farmers
- ❖ Place Order
- ❖ Delivery coordination
- ❖ Price transparency

SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

The **Software Development Life Cycle (SDLC)** is a structured process used to develop software systems from the initial idea to deployment and maintenance. For the **Farm-Yad Project**, SDLC provides a systematic approach to building a platform that connects rural farmers with urban buyers and delivery drivers.

Source: copilot



Phase 1: Planning & Scope Definition

- ❖ **Activities:** Define the vision of Farmyad, identify core constraints (such as low internet bandwidth in provinces), and outline the MVP (Minimum Viable Product) features.
- ❖ **Deliverables:** Project charter, stakeholder identification matrix, and high-level release plan.

Phase 2: Requirements Analysis

- ❖ **Activities:** Collect functional and non-functional requirements by interviewing farmers in the provinces and local transport drivers. This information is drafted and compiled into a Product Backlog.
- ❖ **Key User Stories:** As a farmer, I want to list my harvested cassava crop so that people in the big city can buy wholesale and sell to buyers or personal use. And they can view my inventory.

As a logistics driver, I want to view optimized transport routes so that I can collect produce efficiently from multiple small farms.

Phase 3: Architecture & Software Design

- ❖ **Activities:** Design the database structures, define API contracts, and create user interface mockups.
- ❖ **Deliverables:** Entity-Relationship Diagrams (ERD), UI/UX wireframes in Figma, and system architecture topology.

Phase 4: Iterative Development & Sprints

- ❖ **Activities:** The project is split into 2-week sprints. Developers write clean code, manage source control via GitHub, and perform continuous integration testing (unit tests for inventory matching and route computation algorithms).

Phase 5: Deployment & Release

- ❖ **Activities:** Deploy the backend application to cloud infrastructure and publish the Android mobile application package (APK) to a public registry optimized for low-bandwidth distribution.

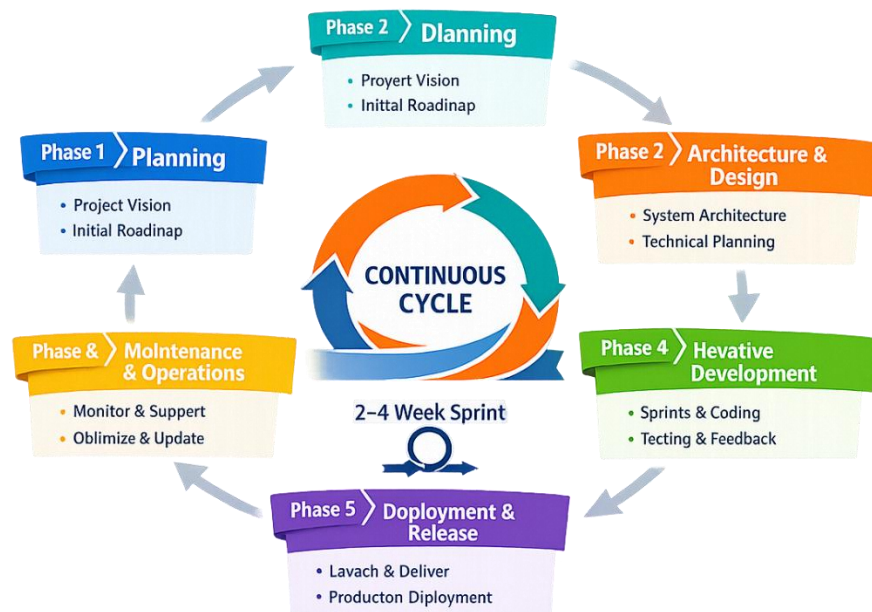
Phase 6: Maintenance & Operations

- ❖ **Activities:** Monitor server performance metrics, patch application vulnerabilities, and optimize data caching to minimize cellular data costs for rural users.

Agile Scrum Methodology.

Agile is ideal for this public-facing application because it allows for iterative development, frequent testing, and rapid adaptation based on direct feedback from rural farmers and logistics operators.

Agile Project Lifecycle



System Architecture Design

The Farmyard platform uses a modular **Three-Tier Architecture** to guarantee high availability, easy maintenance, and reliable processing.

1. PRESENTATION LAYER

- ❖ **Farmer & Driver Mobile Application:** A lightweight, cross-platform app designed for offline-first performance, allowing data synchronization when a cell signal becomes available.
- ❖ **Urban Buyer & Admin Web Dashboard:** A portal for bulk buyers to manage orders, track delivery trucks, and view supply analytics.

2. Application Layer (Logic Tier)

- ❖ **API Gateway:** Routes all client traffic securely, manages request throttling, and handles user authentication tokens.
- ❖ **Inventory Service:** Processes available crop types, quantities, pricing updates, and shelf-life tracking.
- ❖ **Logistics & Routing Service:** Computes optimal truck-loading capacities and dispatch routes to minimize fuel consumption.

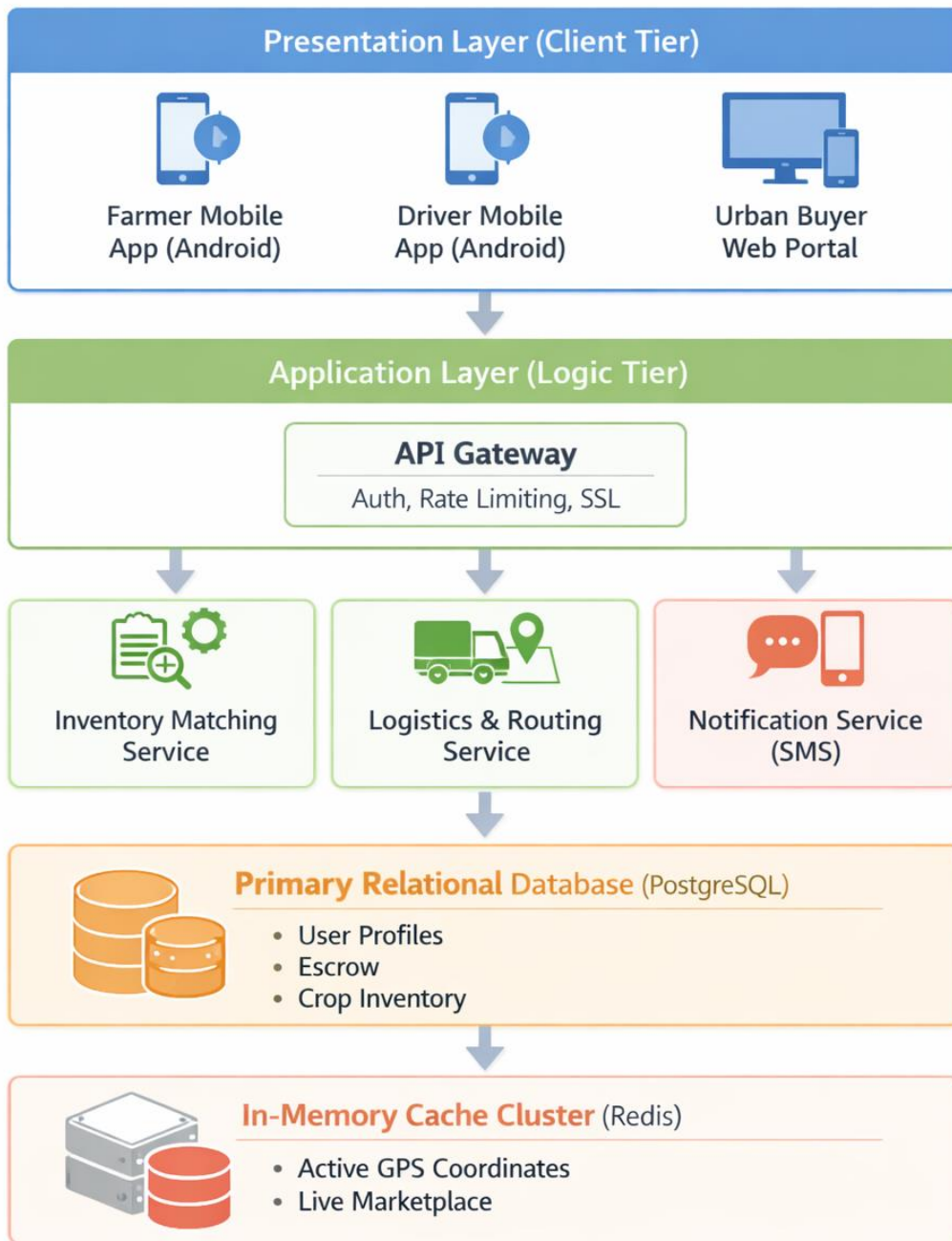
3. Data Layer (Storage Tier)

- ❖ **Relational Database (PostgreSQL):** Stores relational data including user profiles, transactional records, and inventory lists.
- ❖ **Cache Layer:** Caches geographic coordinates and active route updates to ensure responsive performance.

DIAGRAM OF THE ARCHITECTURE

Below is the architectural diagram for the **Farmyard digital public good system**. This structural map fulfills **Task 2 (System Architecture Design)**.

It maps out how data and requests flow from the user applications in the field down to the underlying data storage systems.



Three-Tier Architecture

EXPLAIN HOW COMPONENT INTERACT

PRESENTATION LAYER

FARMER MOBIL APP

It provides a user friendly interface that allows farmers to interact with the system using their mobile phones. Through the app, farmers can register and manage their accounts, upload details of their accounts, upload details of their farm produce, view order from buyer, track deliveries, receive notifications, and communicate with customers.

Driver Mobile App

This enables drivers to interact with the FarmYad digital public good system through their mobile devices. The app allows drivers to view and accept delivery requests, access pickup and delivery locations, track routes, update delivery status, and communicate with farmers and buyers when necessary. It helps ensure the efficient transportation of farm produce from the province areas to customers, improving delivery speed and reducing logistical challenges within the system.

Urban Buyer Web Portal

This is a Web-based interface that allows buyers in cities to access the FarmYad digital public good system through a computer or mobile browser. The portal enables buyers to create accounts, browse available farm products, place orders, make payments, track deliveries, and communicate with farmers when needed. It provides a convenient and secure platform for urban customers to purchase fresh agricultural products directly from farmers, helping to reduce middlemen and improve access to quality produce.

Inventory Matching Service

The Inventory Matching Service is a component of the application layer that automatically matches buyers' orders with the available farm produce listed by farmers. It checks product availability, quantity, quality, and location to identify the most suitable farmer for each order. This service helps ensure efficient order processing, reduces delays, prevents overselling of products, and improves the overall accuracy and reliability of transactions within the farmYad marketplace system.

Logistics and Routing Service

The Logistics and Routing Service is responsible for managing the transportation and delivery of farm produce from farmers to buyers. It helps identify available drivers, assign deliveries, and determine the most efficient routes to reduce transportation costs and delivery time.

Functions:

- ❖ Matches delivery requests with available drivers.
- ❖ Plans the shortest and most efficient delivery routes.
- ❖ Tracks the movement of goods during transit.
- ❖ Provides delivery status updates to farmers and buyers.
- ❖ Helps ensure products reach their destination safely and on time.

In the FarmYad digital public good system, this service acts as the link between farmers in provinces and buyers in big cities like Freetown by coordinating the movement of goods efficiently.

The Notification Service receives information from other application services, such as order management, inventory matching, payment processing, and logistics services, and sends relevant SMS alerts to farmers, buyers, and drivers. It ensures users are

Notification Service (SMS)

Within the Application Layer, the Notification Service (SMS) handles all communication-related functions by generating and sending SMS notifications to users based on events occurring in the system.

informed about order placements, payment confirmations, delivery updates, and other important activities.

Role in the Application Layer:

- ❖ Processes notification requests from other services.
- ❖ Generates appropriate SMS messages.
- ❖ Sends real-time alerts to users.
- ❖ Maintains communication between system stakeholders.
- ❖ Improves user engagement and service reliability.

This service supports seamless communication across the mobile app by ensuring that important information reaches users promptly.

UML DIAGRAMS

The **UML (Unified Modeling Language) Diagram** is used to visually show how the system works, the users involved, and how different parts of the system interact. For **Fam-Yad digital public good System**,

Brief Explanation of the UML Diagram

The UML Diagram for the Fam-Yad system represents the relationships between the main users of the platform and the system functions. The primary actors are **Farmers, Buyers, Drivers, and Admins**. Farmers can register, upload products, manage listings, and receive order notifications. Buyers can browse products, place orders, make payments, track deliveries, and leave reviews. Drivers handle product transportation and

update delivery status, while Admins manage users, monitor orders, verify products, and resolve issues.

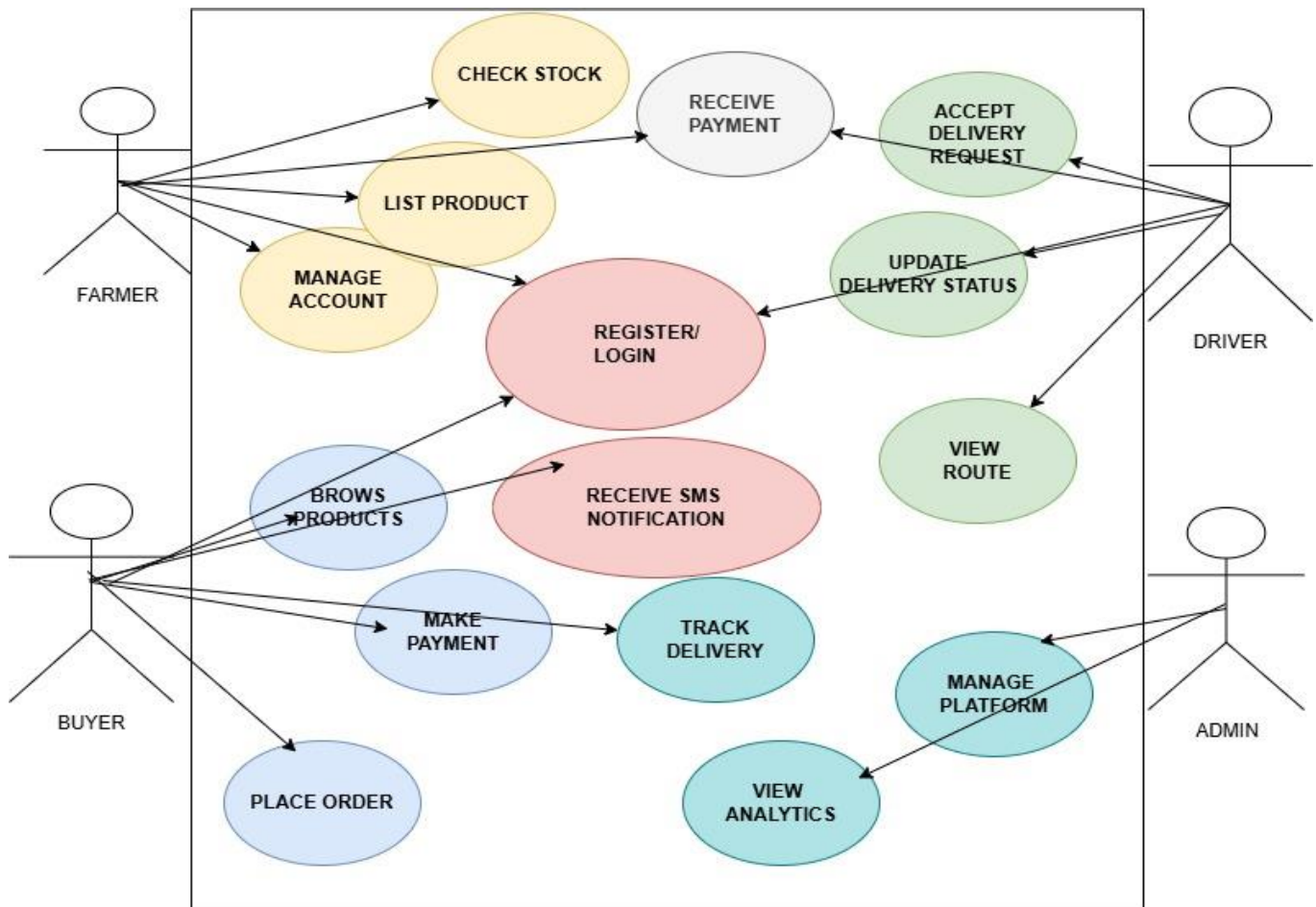
The UML Diagram helps stakeholders understand the system structure before development begins. It clearly illustrates the flow of activities from product listing by farmers, to purchasing by buyers, payment processing, delivery coordination, and feedback collection. This makes the system easier to design, develop, and maintain.

Main Use Cases in the UML Diagram

- ❖ User Registration
- ❖ User Login
- ❖ Upload Product
- ❖ Browse Products
- ❖ Add to Cart
- ❖ Place Order
- ❖ Make Payment
- ❖ Track Delivery
- ❖ Receive Notifications
- ❖ Rate and Review Products
- ❖ Manage Users and Orders (Admin)

This UML Diagram provides a visual blueprint of how the Fam-Yad platform connects upline farmers (like Bo) with Big city buyers (like Freetown) through a structured digital marketplace.

USECASE DIAGRAM



Farm-Yad system, "View Analytics" could allow farmers and administrators to see information such as:

- ❖ Total products sold
- ❖ Total orders received
- ❖ Revenue earned
- ❖ Most popular products
- ❖ Number of registered farmers and buyers

- ❖ Delivery success rates
- ❖ Monthly sales trends

Example

A farmer clicks "**View Analytics**" and sees:

- ❖ Products Sold: 120 kg of rice
- ❖ Orders Completed: 35
- ❖ Revenue Earned: Le 15,000
- ❖ Best-Selling Product: Tomatoes

An admin might see:

- ❖ Total Farmers: 500
- ❖ Total Buyers: 2,000
- ❖ Total Orders This Month: 1,500

Brief Definition

View Analytics: A feature that allows users to view reports, statistics, and insights about system activities and performance to support better decision-making.

ACTIVITY DIAGRAM

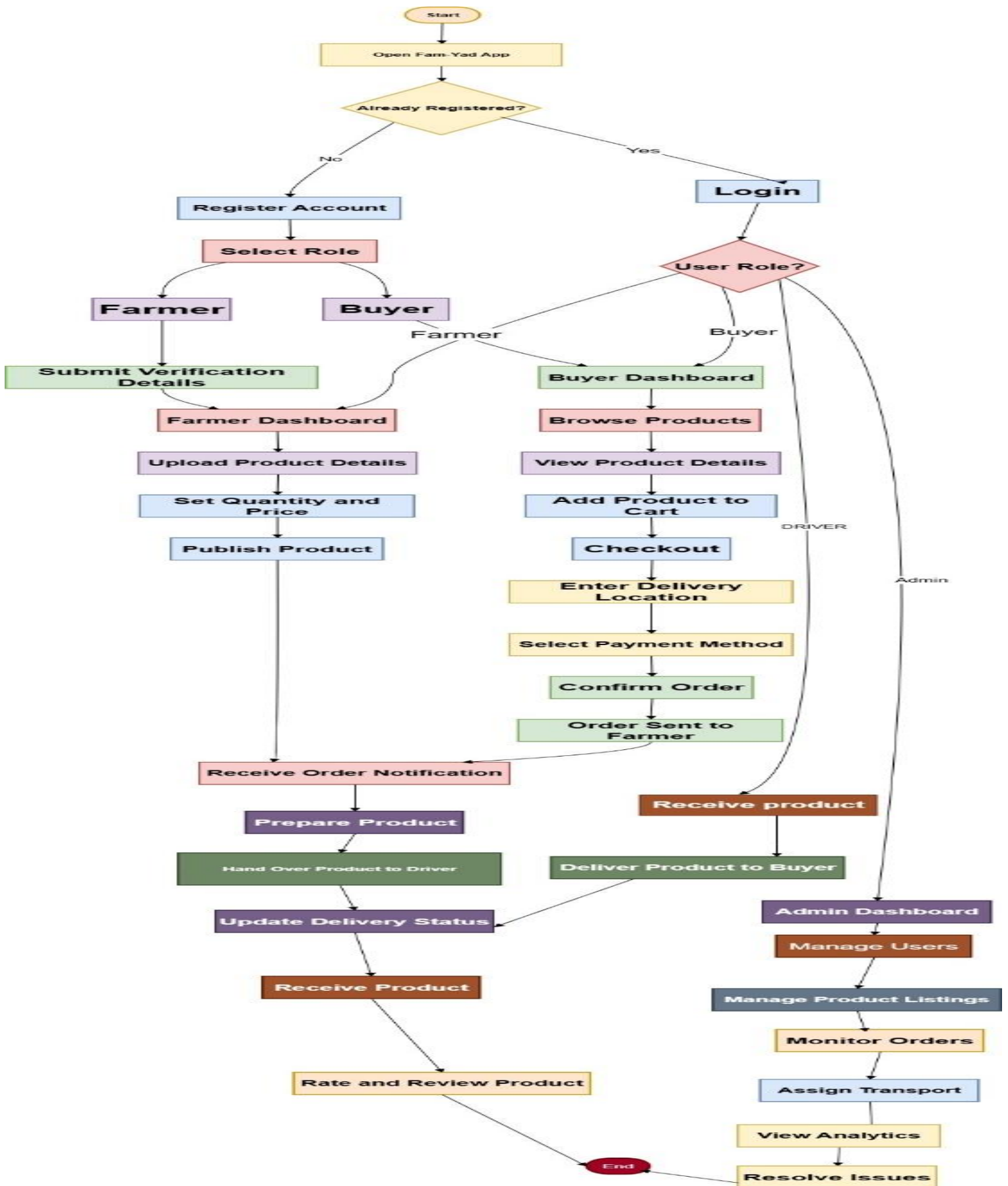
The **Farm-Yad** activity diagram shows the journey of users as they interact with the platform. It provides a clear picture of how activities flow from one step to another, starting when a user opens the application and ending when a transaction is successfully completed.

The process begins with a user registering or logging into the system. Farmers can create accounts, upload information about their farm produce, set prices, and receive notifications whenever buyers place orders. Buyers can browse available products, compare prices, add items to their cart, choose a payment method, and place orders. Once an order is confirmed, the farmer prepares the goods and hands them over to a driver for delivery.

The driver then transports the products and updates the delivery status so that buyers can track their orders in real time. After receiving the products, buyers can leave ratings and reviews to help build trust within the platform. They can also view analytics such as order history and purchasing trends. Meanwhile, administrators oversee the entire process by managing users, monitoring orders, assigning transportation, and resolving any issues that may arise.

Overall, the Activity Diagram helps explain how Farm-Yad connects province farmers with City buyers through a simple and organized workflow, ensuring that products move more efficiently from farms to customers while improving market access and reducing food waste.

Activity Diagram



CLASS DIAGRAM

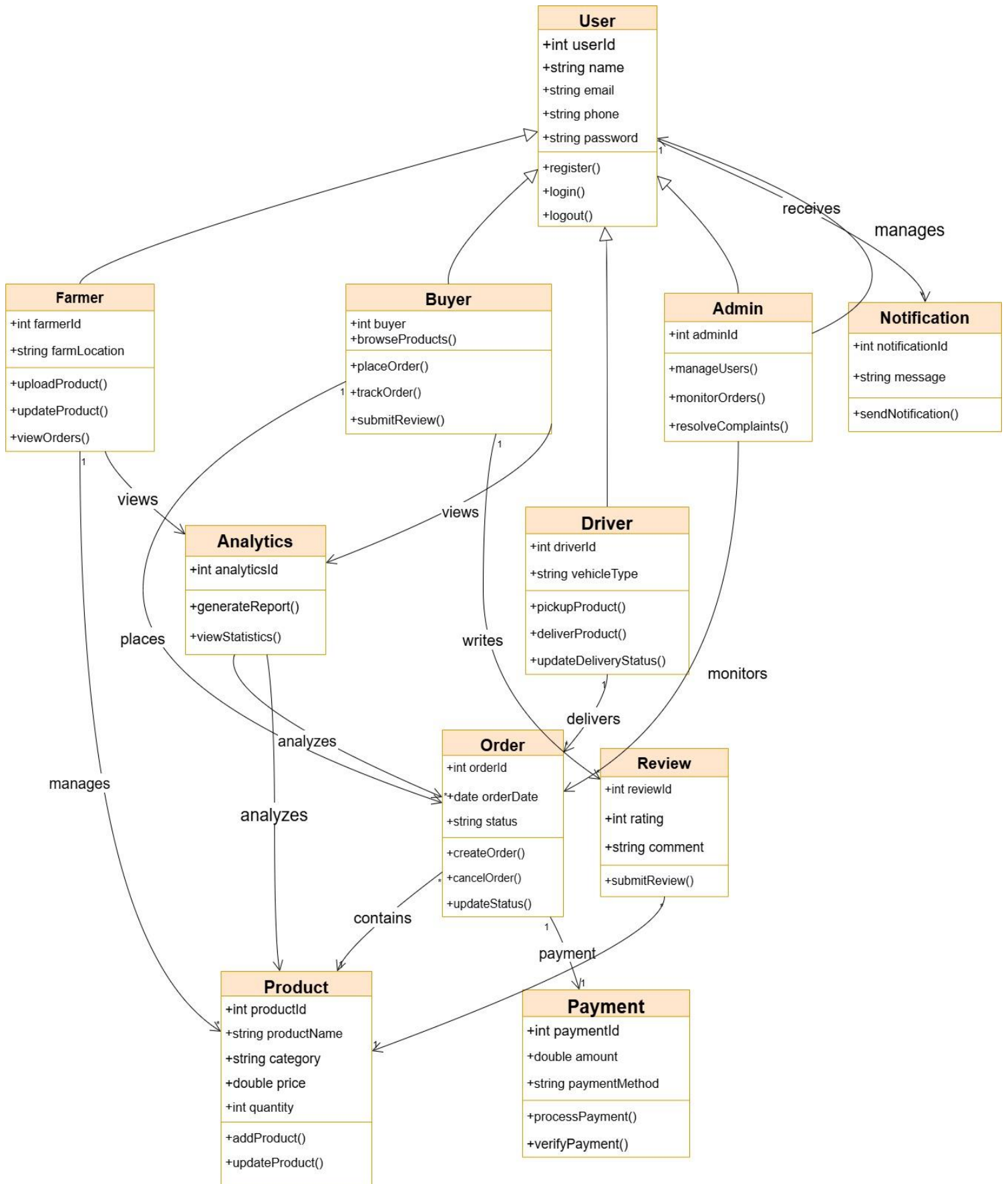
The **Class Diagram** for the **Farm-Yad digital public good System** provides a visual representation of the main components of the system and how they are connected. It helps us understand the structure of the application by showing the different classes, their attributes, functions, and relationships.

In the Farm-Yad system, the **User** class serves as the foundation because every person using the platform must first create an account and log in. From this class, other specialized classes such as **Farmer**, **Buyer**, **Driver**, and **Admin** are derived. Each of these users has different responsibilities within the system.

The **Farmer** class is responsible for uploading and managing farm products, while the **Buyer** class allows customers to browse products, place orders, make payments, and leave reviews. The **Driver** class handles the transportation of products from farmers to buyers and updates delivery statuses. The **Admin** class oversees the entire platform by managing users, monitoring orders, and resolving issues.

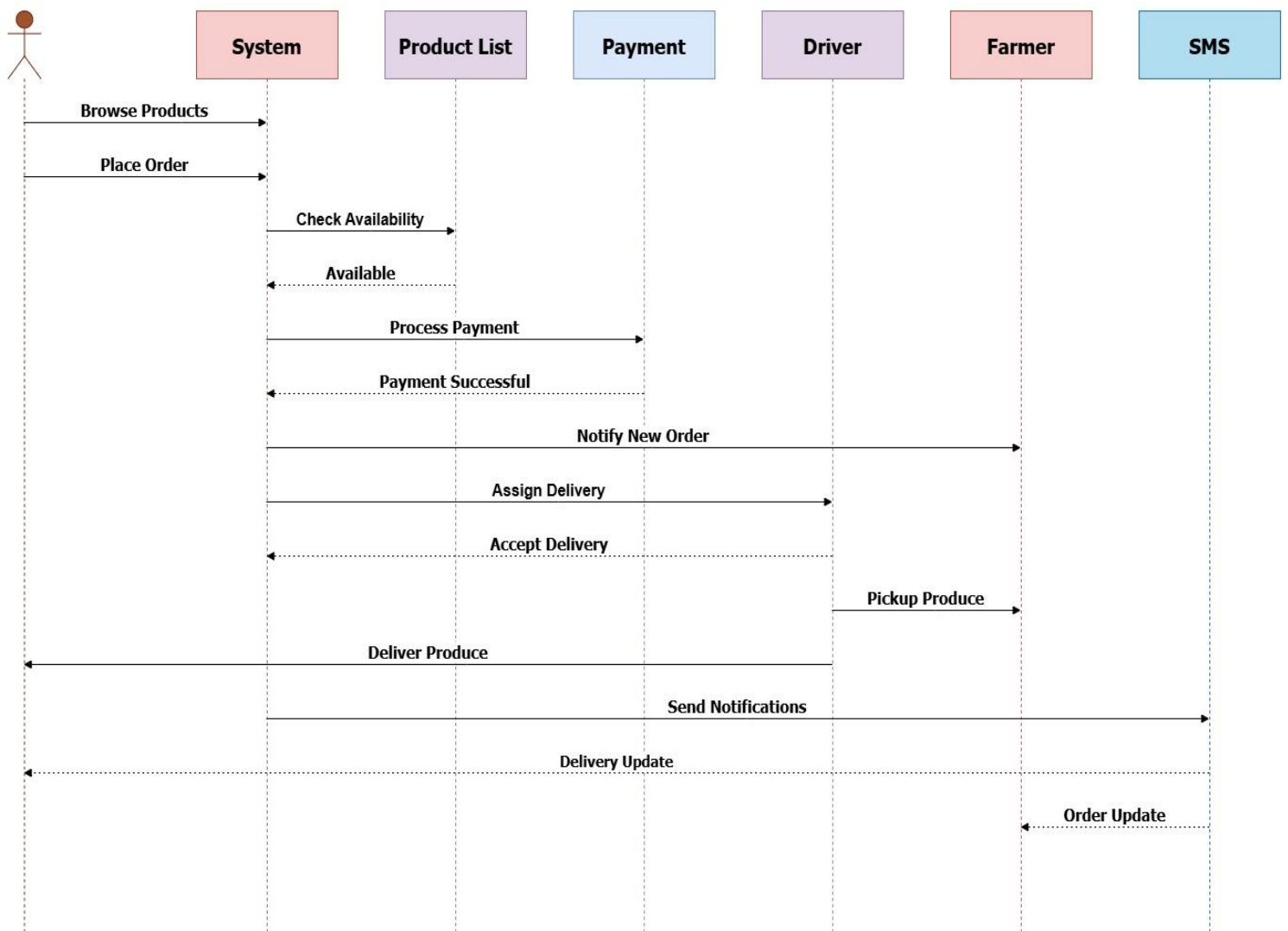
Other important classes include **Product**, which stores information about farm produce; **Order**, which manages purchase transactions; **Payment**, which handles payment processing; **Review**, which stores customer feedback; **Notification**, which sends updates to users; and **Analytics**, which generates reports and statistics about system activities.

Overall, the Class Diagram acts as a blueprint for the Farm-Yad system. It clearly shows how different parts of the platform work together to connect farmers with buyers, manage transactions, coordinate deliveries, and ensure the smooth operation of the agricultural marketplace. This makes it easier for developers, stakeholders, and project supervisors to understand the design and functionality of the system before implementation begins.



SEQUENCE DIAGRAM

The Sequence Diagram shows how different users interact with the Fam-Yad system over time. The process starts when farmers and buyers register or log into the platform. Farmers upload their products, while buyers browse available products and place orders. The system processes payments and notifies farmers of new orders. After the farmer accepts the order, a driver collects the products and delivers them to the buyer. Throughout the process, the system updates delivery information and allows buyers to track their orders. Once delivery is completed, buyers can leave ratings and reviews.



Administrators monitor activities, manage users, and resolve issues to ensure the platform operates smoothly. This diagram helps visualize the order in which messages and actions occur between users and the system.

SUSTAINABLE DEVELOPMENT GOALS ALIGNMENT

This system supports **Sustainable Development Goal 2: Zero Hunger** by improving food distribution, reducing waste, and making quality food more accessible to urban population while increasing income for rural farmers. According to the Food and Agriculture Organization (FAO), a significant portion of food produced in developing countries is lost due to poor transportation, storage, and market access, while urban populations still face limited access to affordable, high-quality food. By connecting farmers directly to buyers and coordinating delivery, the platform reduces post-harvest losses, increases food availability in cities, and helps farmers earn better income, thereby strengthening food security. This aligns with **SDG 2's target** of ensuring access to safe, nutritious, and sufficient food for all while promoting sustainable agricultural systems, as outlined by the United Nations in its Sustainable Development Goals Framework; 2030. This project also strongly supports **Sustainable Development Goal 8**, Creating economic opportunities and **Sustainable Development Goal 9** Promoting digital innovation in agriculture.

Network Definition

A network is a visual representation of the project activities and their dependencies

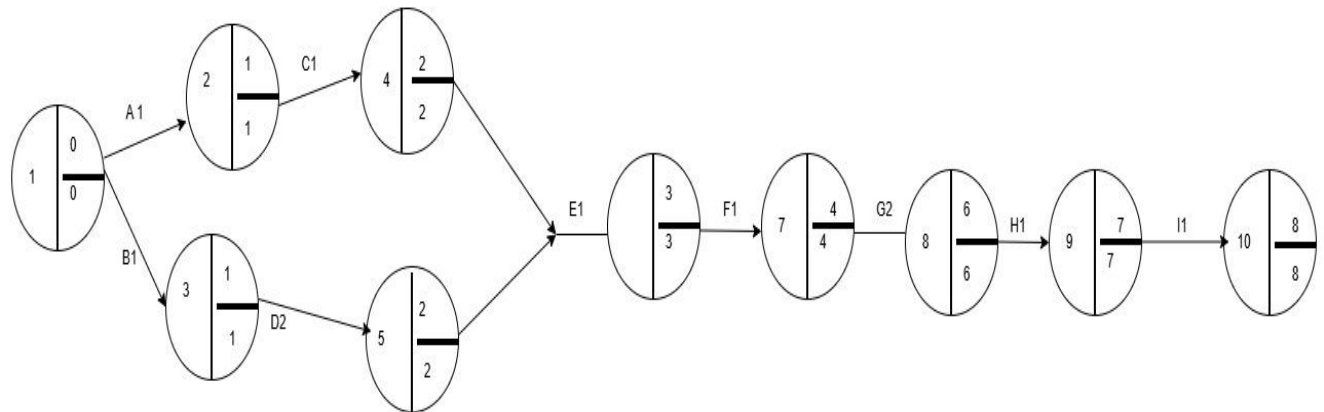
PROJECT ACTIVITY SCHEDULE (TIMELINE TABLE)

Week	Activity
Week 4	Project Planning & Requirements Gathering
Week 5	System Analysis
Week 6	System Design (UML Diagrams)
Week 7	UI/UX Design (Figma Prototype)
Week 8	System Development – Core Modules
Week 9	System Development – Remaining Modules
Week 10	Testing & Bug Fixing
Week 11	Documentation, Deployment & Final Review
Week 12	Project Presentation

PRECEDENCE TABLE

	ATTRIBUTE	PREDECESSOR	DURATION
P	A	–	1
SA	B	–	1
SD	C	A	1
D	D	B	2
SDC	E	C,D	1
SDR	F	E	1
T	G	F	2
DD	H	G	1
PP	I	H	1

NETWORK DIAGRAM



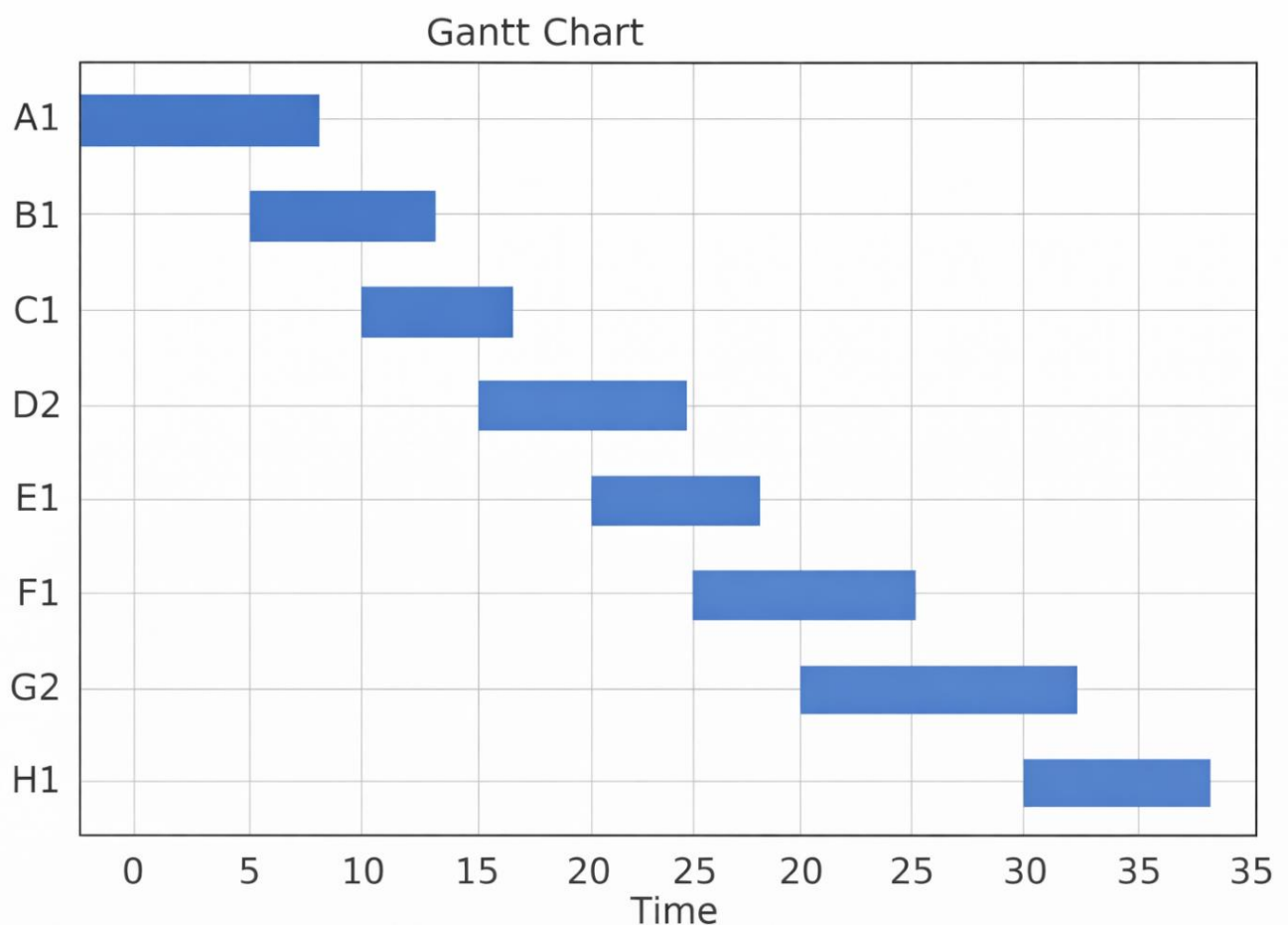
Network Diagram Explanation

The Network Diagram shows the logical sequence and dependencies between the various activities involved in the development of the Farm-Yad digital public good System. It shows how each project phase depends on the successful completion of the previous phase. The project starts with planning and requirements gathering, followed by system analysis, system design, UI/UX design (Figma prototype), system development core modules, system development remaining modules, testing and bug fixing, documentation deployment and final review and project presentation. The diagram helps identify the workflow of the project and ensures that tasks are carried out in the correct order. It also assists in project scheduling and resource allocation by highlighting the critical path of project activities.

Gantt Chart

The Gantt Chart provides a visual representation of the project schedule for the Fam-Yad Agricultural Marketplace System. It shows the sequence of activities, their duration, and the timeframe within which each task is expected to be completed. The project begins in Week 4 with project planning and requirements gathering and progresses through system analysis, design, development, testing, deployment, and documentation. All project

activities are scheduled to be completed by Week 11, allowing Week 12 to be reserved for project presentation and demonstration. The Gantt Chart helps the project team monitor progress, manage time effectively, and ensure that all project milestones are achieved within the specified period.



The diagram is a **visual schedule** showing how project tasks (A1 through I1) progress over time.

- The **Y-axis** lists each activity in sequence, while the **X-axis** represents time units.
 - Each **blue bar** corresponds to a task's duration and start–end times.
 - Tasks are arranged so that dependent activities begin only after their predecessor's finish — for example, task **C1** starts after **A1**, and **E1** begins once **C1** and **D2** are complete.
 - The chart helps identify the **workflow order**, overlaps, and idle periods, making it easy to track progress and manage deadlines.
-

PROTOTYPE

What is a Prototype?

A **prototype** is an early or preliminary version of a product, system, or application that is created before the final version is developed. It is used to test ideas, explore different design options, and identify any problems before investing time and resources in full development.

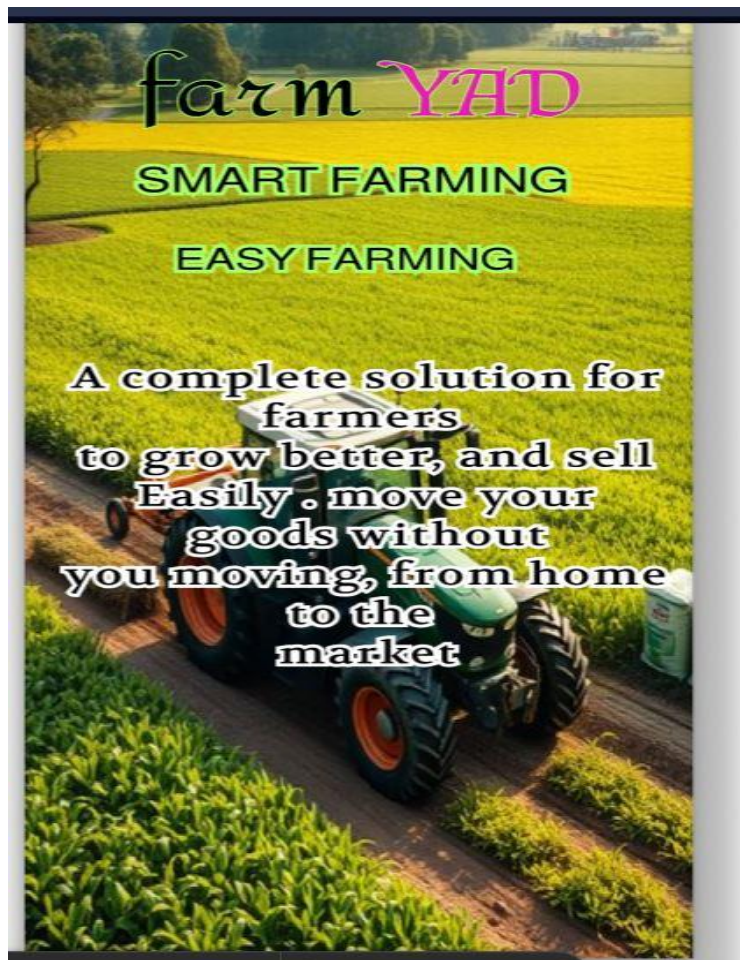
Instead of building the complete product immediately, developers first create a prototype to see how the system will work and how users will interact with it. This allows designers, developers, and stakeholders to gather feedback, make improvements, and ensure the final product meets user needs and project requirements.

For example, when developing a mobile app, a prototype may consist of simple screens that demonstrate how users will navigate through the application. Although it may not include all the final features, it provides a clear representation of the system's layout, functionality, and user experience.

Using a prototype helps reduce development risks, saves time and costs by detecting issues early, improves communication among project team members, and increases the chances of delivering a successful product that satisfies users.

Below is my screen for the FarmYad Digital Public Good System

SCREEN 1 (Splash Screen)



This screen is the **splash (welcome) screen** of the **Farm YAD** application. It is the first screen users see when they open the app, serving as an introduction to the platform and its purpose.

The background image of a green farmland and tractor represents modern agriculture and immediately communicates that the application is designed for farming activities. At the top, the **Farm YAD** logo establishes the application's brand identity, while the slogans

"Smart Farming" and **"Easy Farming"** emphasize the app's goal of making farming more efficient through technology.

The message displayed on the screen explains the main purpose of the application: to provide farmers with a complete solution for improving agricultural production and marketing. It highlights that farmers can grow crops more effectively, connect with buyers, and transport or sell their produce without having to travel from their homes to the market. This reflects the application's focus on convenience, accessibility, and reducing the challenges farmers face in selling their products.

Overall, the splash screen creates a positive first impression by introducing the Farm YAD brand, communicating its mission, and preparing users to access the application's features.

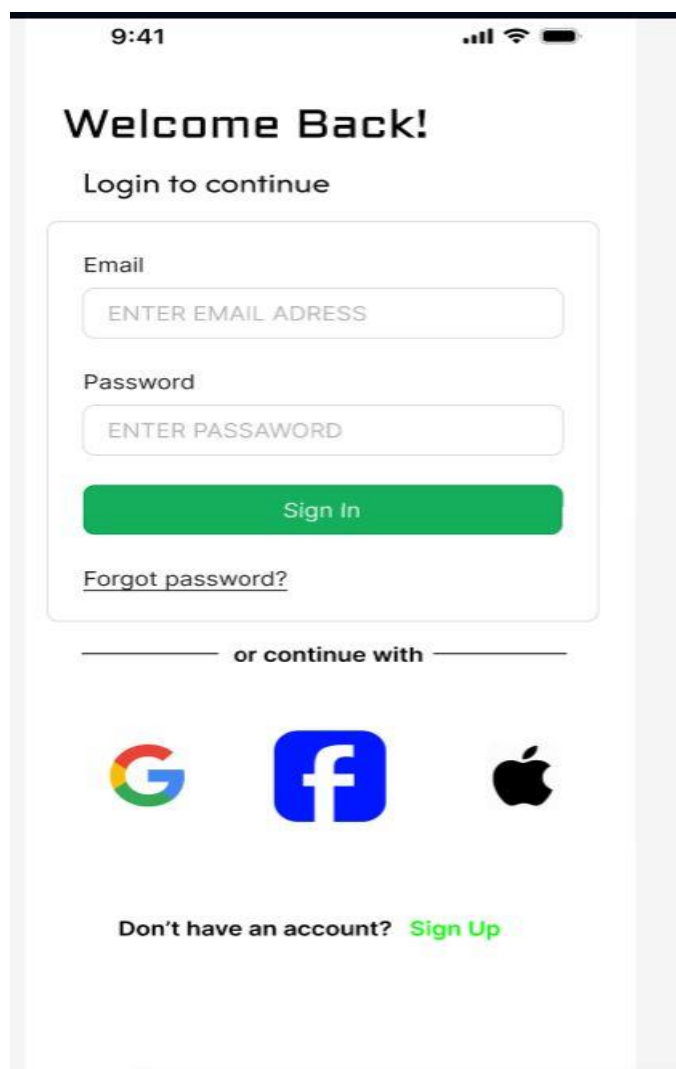
SCREEN 2(Welcome Page)



The **Welcome Screen** is the first interface users see after the splash screen. It introduces users to the Farm YAD application and highlights its main purpose. The screen displays the Farm YAD logo and a farming-themed illustration to reinforce the

application's focus on agriculture. A short description explains that the platform allows farmers to sell their produce, export agricultural products, and connect with customers digitally across Sierra Leone. The **"Get Started"** button directs new users to the registration process, while the **"Login"** option allows existing users to access their accounts. This screen serves as the entry point to the application and encourages users to begin using its services.

SCREEN 3(Log in)

A mobile application login screen. At the top, the status bar shows the time 9:41, signal strength, Wi-Fi, and battery. The main heading is "Welcome Back!" followed by the instruction "Login to continue". Below this is a white rounded rectangle containing the login form. It has an "Email" label above a text input field with the placeholder "ENTER EMAIL ADDRESS". Below that is a "Password" label above another text input field with the placeholder "ENTER PASSAWORD". A green "Sign In" button is positioned below the password field. A link for "Forgot password?" is at the bottom of the form. Below the form, the text "or continue with" is centered between two horizontal lines. Underneath are three social media icons: Google, Facebook, and Apple. At the very bottom, the text "Don't have an account?" is followed by a green "Sign Up" link.

9:41

Welcome Back!

Login to continue

Email

Password

Sign In

[Forgot password?](#)

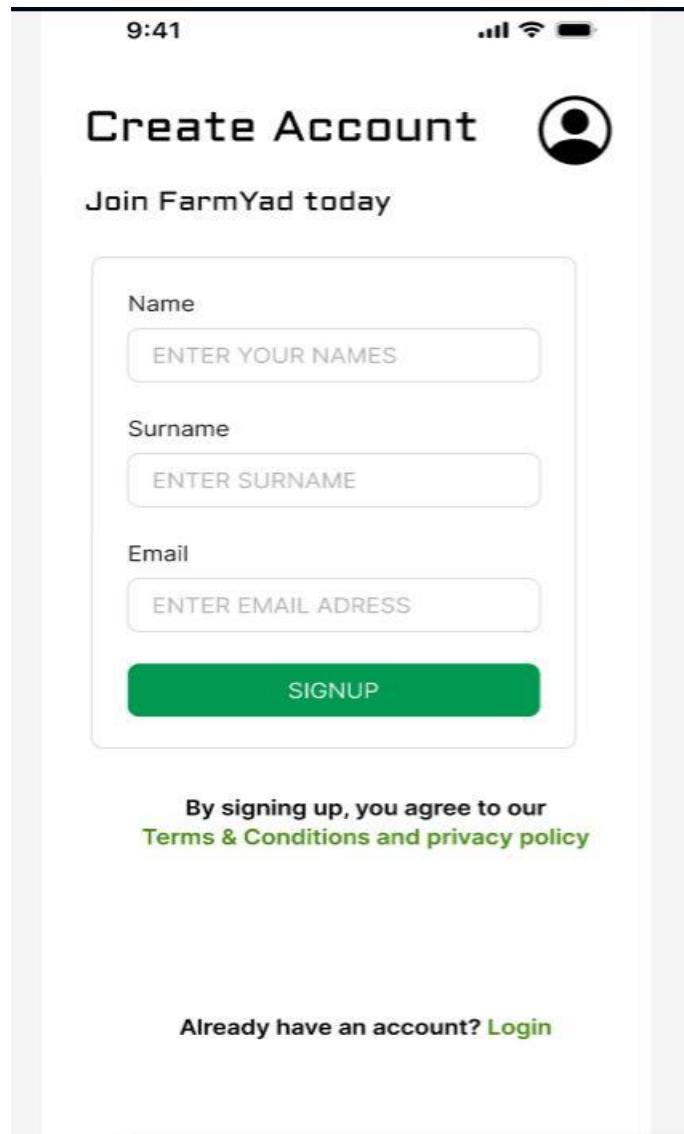
or continue with

Don't have an account? [Sign Up](#)

The **Login Screen** allows registered users to securely access their Farm YAD accounts. Users are required to enter their **email address** and **password**, then tap the "**Sign in**" button to log into the application. The screen also provides a "**Forgot Password?**" option, enabling users to recover their accounts if they cannot remember their password. For added convenience, users can choose to sign in using their **Google, Facebook**, or **Apple** accounts. At the bottom of the screen, a "**Sign Up**" link is available for new users who do not yet have an account. The simple and intuitive layout makes the login process quick and user-friendly.

SCREEN 4 (Create Account)



9:41

Create Account 

Join FarmYad today

Name
ENTER YOUR NAMES

Surname
ENTER SURNAME

Email
ENTER EMAIL ADDRESS

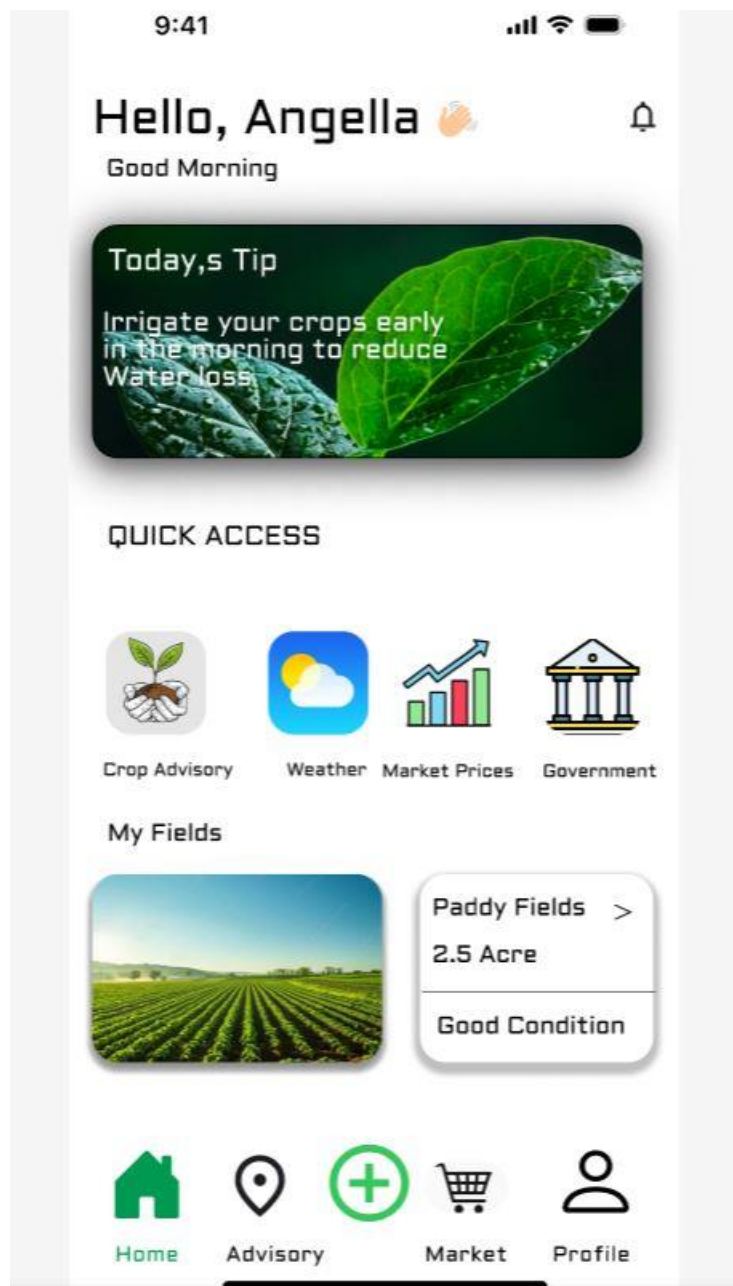
SIGNUP

By signing up, you agree to our
[Terms & Conditions and privacy policy](#)

Already have an account? [Login](#)

The **Create Account Screen** enables new users to register for a Farm YAD account. It contains a simple registration form where users enter their **name**, **surname**, and **email address**. After providing the required information, users can tap the **"Sign Up"** button to create an account. The screen also includes a statement informing users that by registering, they agree to the application's **Terms & Conditions** and **Privacy Policy**, ensuring transparency regarding data usage. At the bottom of the screen, the **"Login"** link is provided for users who already have an existing account. This screen is designed to make the registration process quick, simple, and user-friendly.

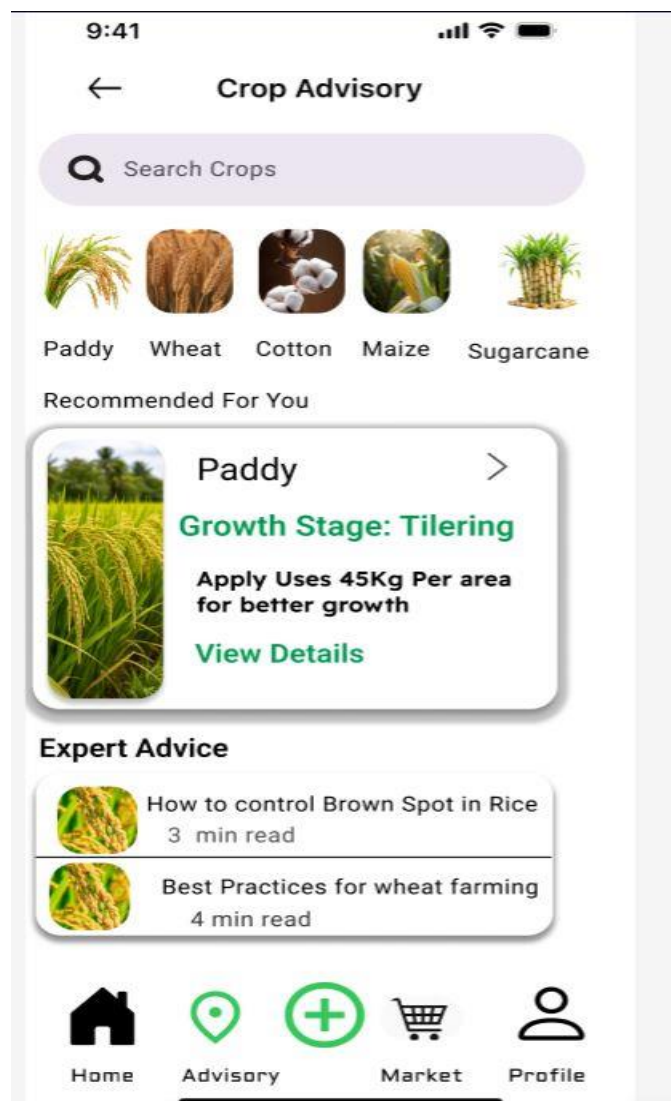
SCREEN 5 (Home Dashboard)



The **Home Dashboard** is the main interface users see after successfully logging into the Farm YAD application. It welcomes the user with a personalized greeting and displays a **daily farming tip** to help improve farming practices. The **Quick Access** section

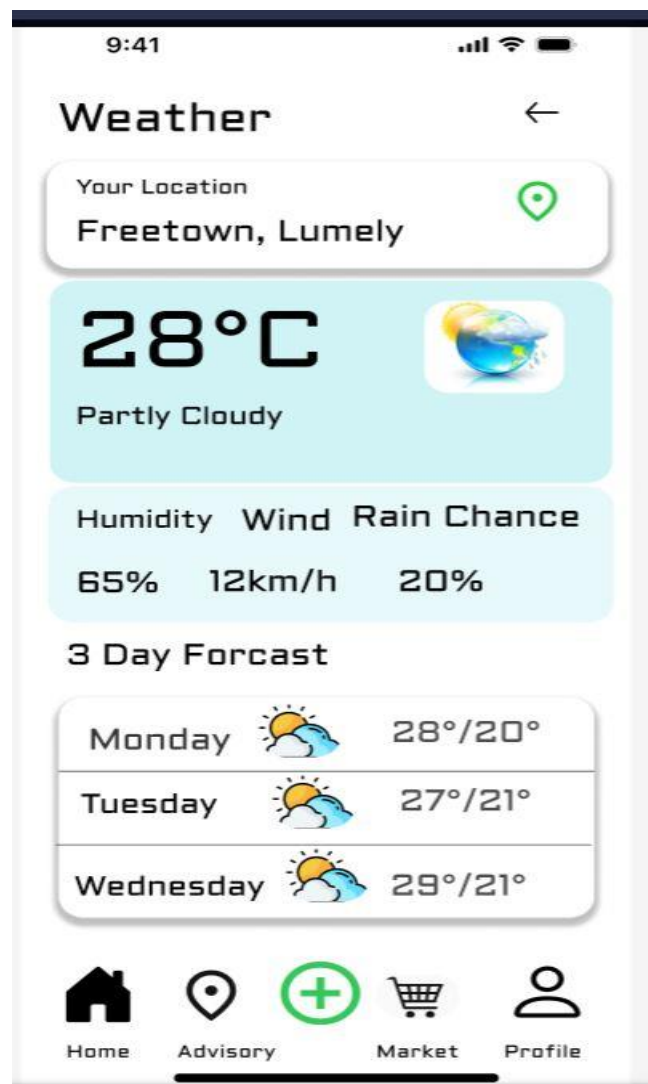
provides shortcuts to important services such as **Crop Advisory**, **Weather Forecast**, **Market Prices**, and **Government** agricultural information. Below this section, the **My Fields** area allows farmers to monitor their farms by displaying details such as the field name, size, and current condition. At the bottom of the screen, the navigation bar enables users to move easily between the **Home**, **Advisory**, **Market**, and **Profile** sections of the application. Overall, the dashboard is designed to give farmers quick access to essential information and tools needed for effective farm management.

SCREEN 6 (Crop Advisory)



The **Crop Advisory** screen provides farmers with expert guidance and recommendations to help them improve crop production and maintain healthy farms. It offers information on the best farming practices, including crop selection, planting schedules, irrigation methods, fertilizer application, pest and disease management, and harvesting techniques. The screen is designed to help farmers make informed decisions based on their crops and farming conditions, leading to increased productivity and reduced crop losses. By providing timely and reliable agricultural advice, the Crop Advisory feature supports farmers in achieving better yields and practicing more sustainable farming.

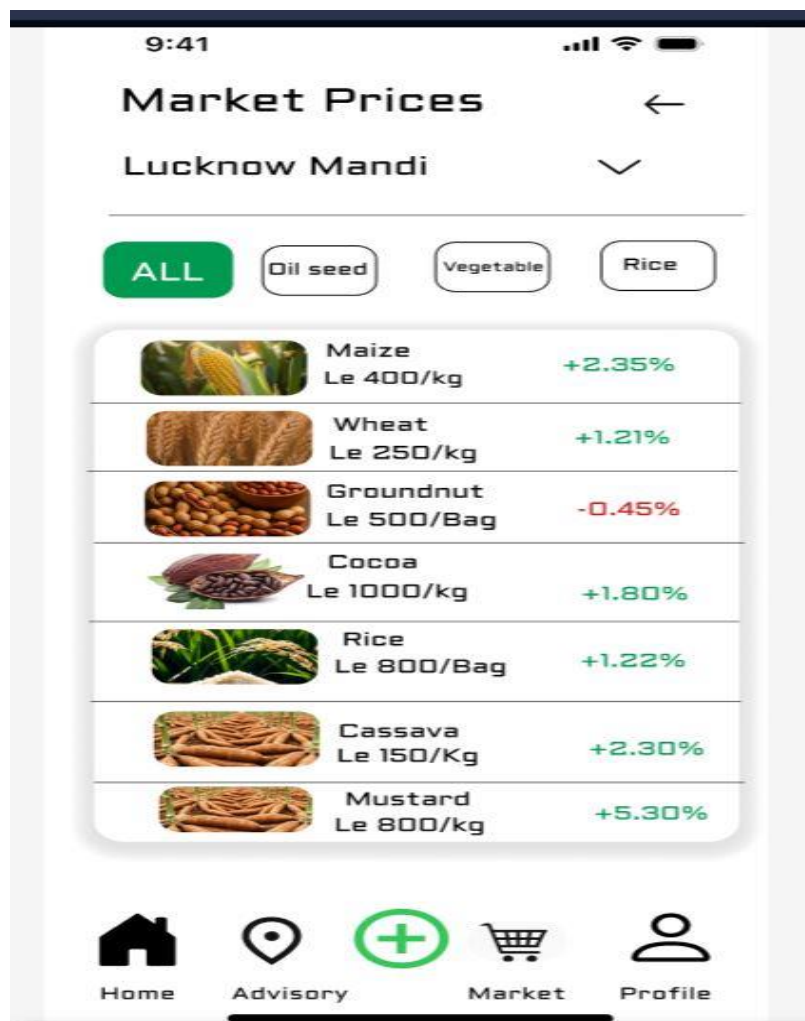
SCREEN 7(Weather)



The **Weather** screen provides farmers with up-to-date weather information to help them plan and manage their farming activities effectively. It displays important weather details such as **temperature, rainfall, humidity, wind speed, and weather forecasts** for the coming days. This information enables farmers to make informed decisions about planting, irrigation, fertilizer application, pesticide spraying, and harvesting.

By providing accurate and timely weather updates, the Weather screen helps farmers reduce the risks associated with unpredictable weather conditions, protect their crops, and improve overall farm productivity. It serves as an essential tool for planning daily farming operations and ensuring better crop management.

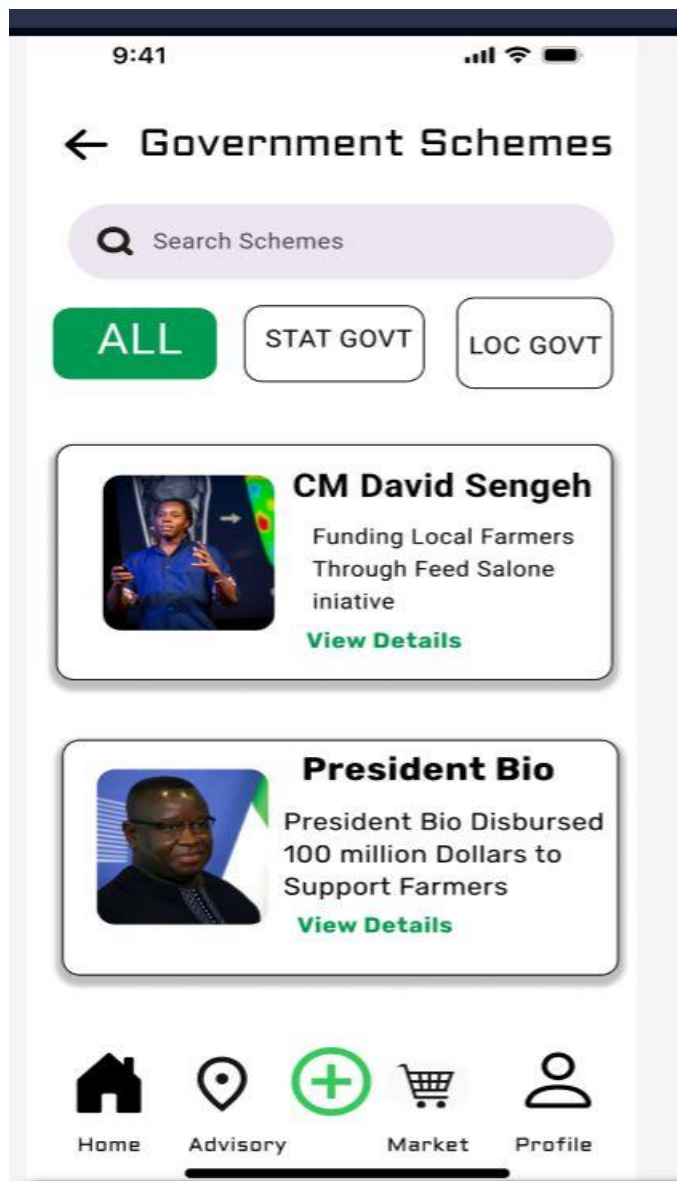
SCREEN 8 (Market Prices)



The **Market Prices** screen provides buyers with current prices of farm products from different farmers. It allows buyers to view the latest market rates for crops and other farm produce, helping them determine the best price of products. farmers can easily sell their product and maximize their profits.

This screen also helps improve transparency between farmers and buyers. With access to real-time or regularly updated price information, farmers can negotiate better prices, plan their sales more effectively, and connect with buyers based on market demand.

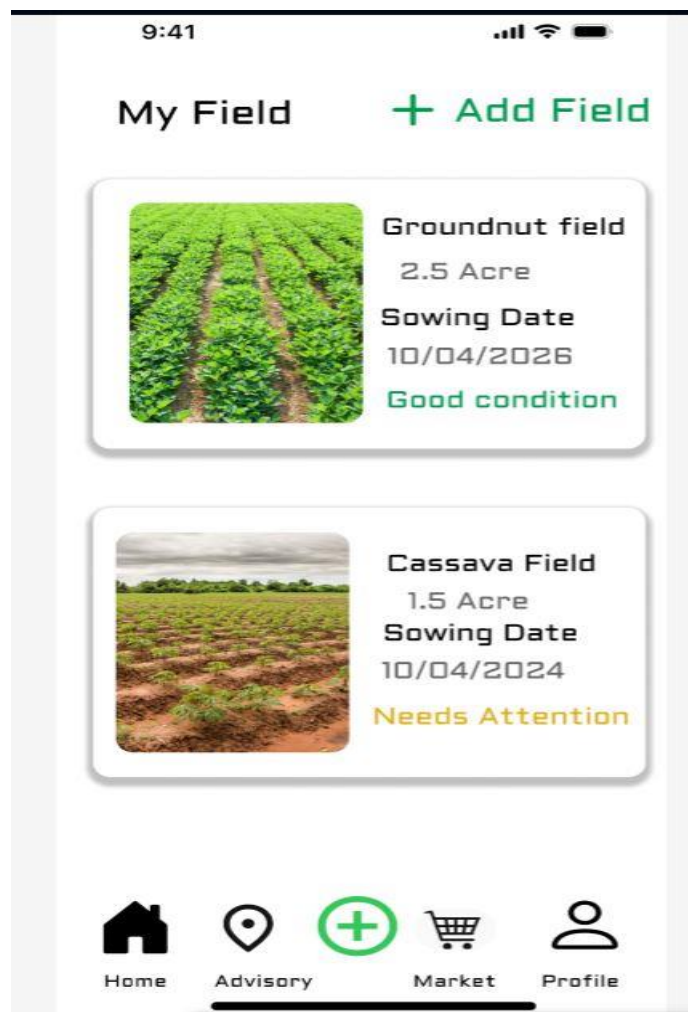
SCREEN 9 (Government Schemes)



The **Government Schemes** screen provides farmers with information about government agricultural programs, subsidies, grants, and support services. It allows users to stay informed about opportunities such as financial assistance, farming loans, training programs, input subsidies, and other initiatives that can improve agricultural productivity and livelihoods.

This screen helps farmers understand the eligibility requirements, application procedures, and deadlines for each scheme, making it easier to access available government support. By providing timely and reliable information, the Government Schemes feature encourages farmers to take advantage of programs that can reduce production costs, improve farming practices, and increase their overall income.

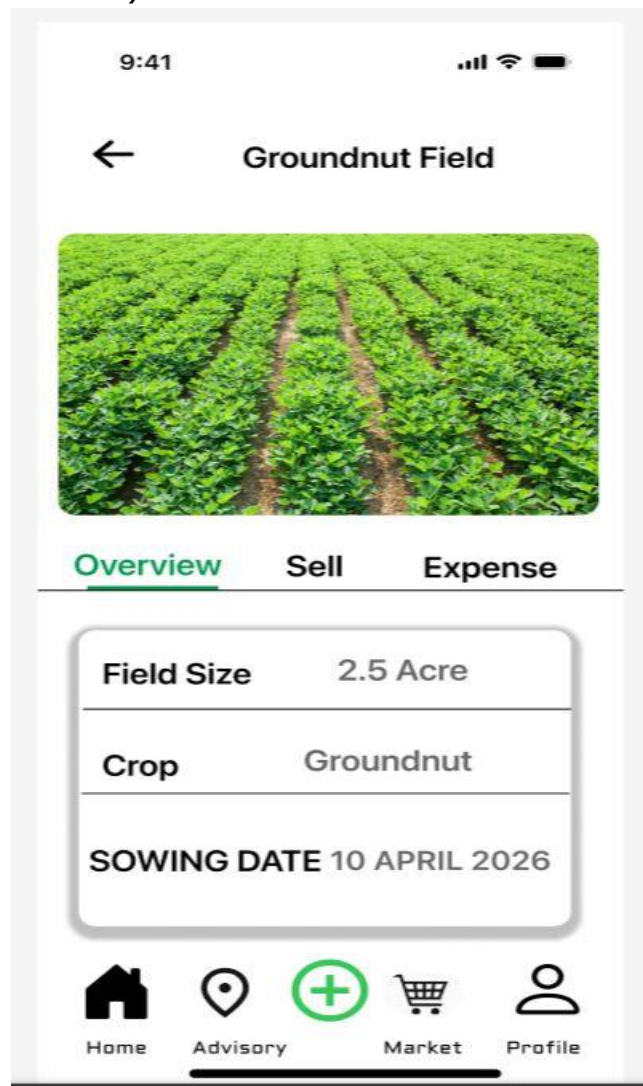
SCREEN 10 (My Field)



The **My Fields** screen allows farmers to manage and monitor all their registered farmland in one place. It displays important information about each field, including the **field name, location, size, crop type, planting date, and current crop condition**. Farmers can use this screen to keep track of their farming activities and monitor the progress of each field throughout the growing season.

The screen also enables users to update field information, record farming activities, and monitor crop performance. By organizing all field-related data in one location, the **My Fields** feature helps farmers make informed decisions, improve farm management, and increase productivity through better planning and record-keeping.

SCREEN 11 (Groundnut field)

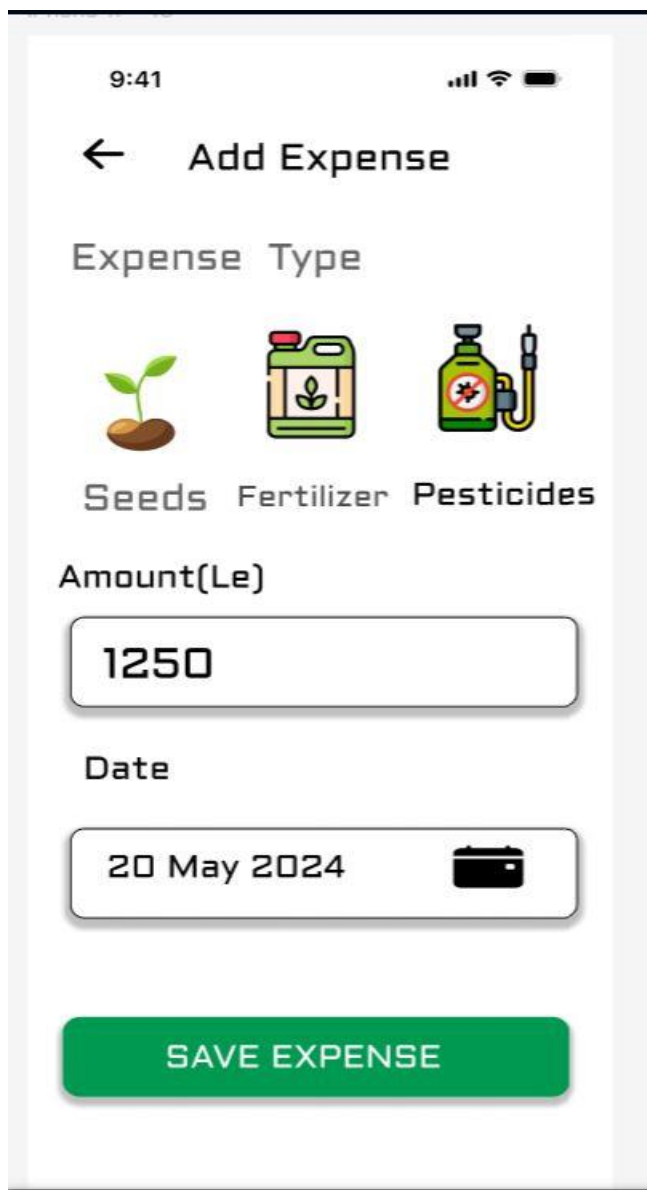


A **groundnut field screen** is a monitoring interface that shows key information about a farmer's groundnut (peanut) farm in one place.

It typically displays details such as crop health, soil condition, growth stage, and field size. It may also include alerts about pests or diseases, recommended farming actions (like weeding or fertilizing), and weather conditions that could affect the crop.

This also help farmers sell their lard farm produce at a go to wholesales buyers.

SCREEN 12 (Add Expense)



9:41

← Add Expense

Expense Type

Seeds Fertilizer Pesticides

Amount(Le)

1250

Date

20 May 2024

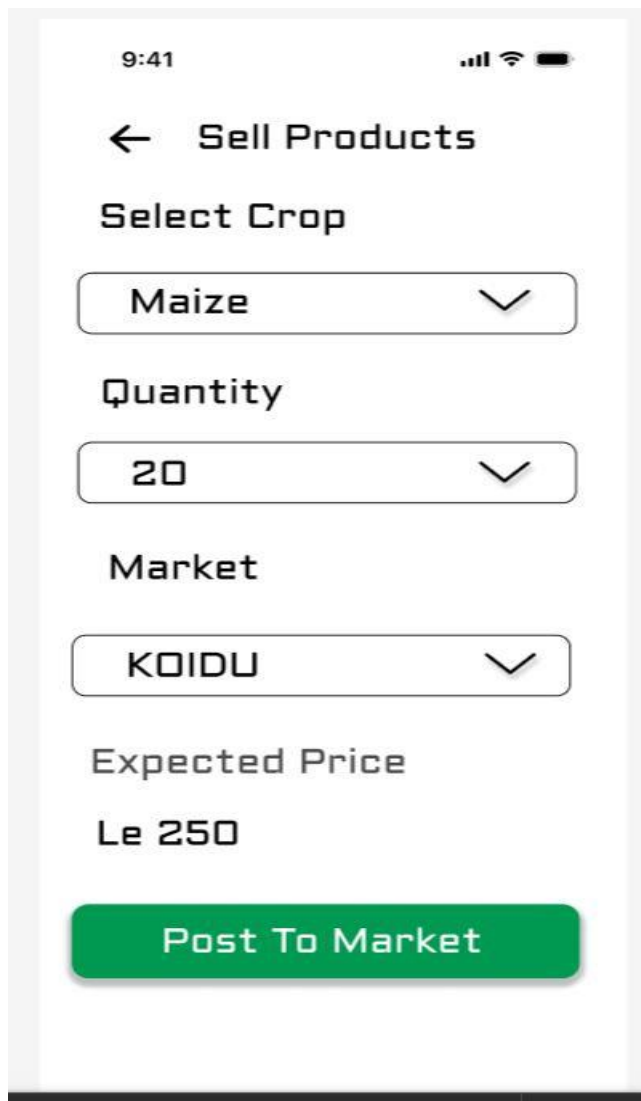
SAVE EXPENSE

The **Add Expense screen** is a section in a farm management system where a user records any money spent on farming activities.

It allows the user to input details such as the expense type (e.g., labor, seeds, fertilizer, transport), amount, date, and a short description. Once saved, the expense is stored for tracking farm costs and budgeting.

Overall, it helps farmers keep accurate financial records and understand how much is being spent on their farming operations.

SCREEN 13 (Sell Product)



9:41

← Sell Products

Select Crop

Maize

Quantity

20

Market

KOIDU

Expected Price

Le 250

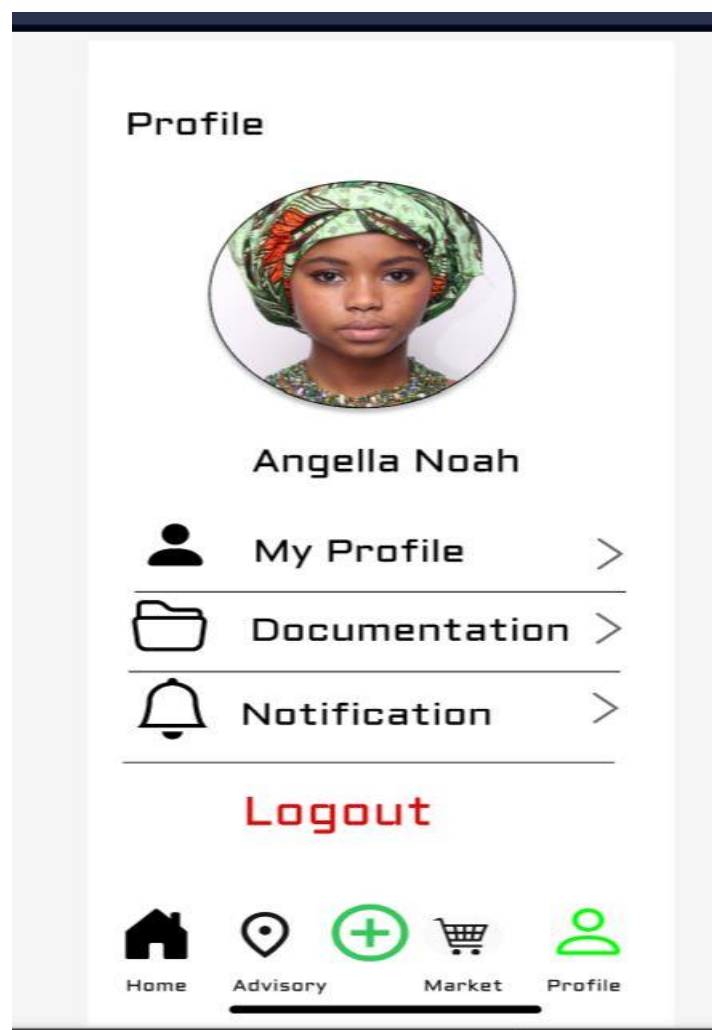
Post To Market

The **Sell Product screen** is where a farmer records or manages the sale of farm produce.

It typically allows the buyers to enter details such as the product name (e.g., groundnuts, maize), quantity, selling price, buyer information, and date of sale. Once submitted, the system updates stock levels and records the income generated.

Overall, it helps farmers track sales, manage inventory, and monitor farm revenue efficiently.

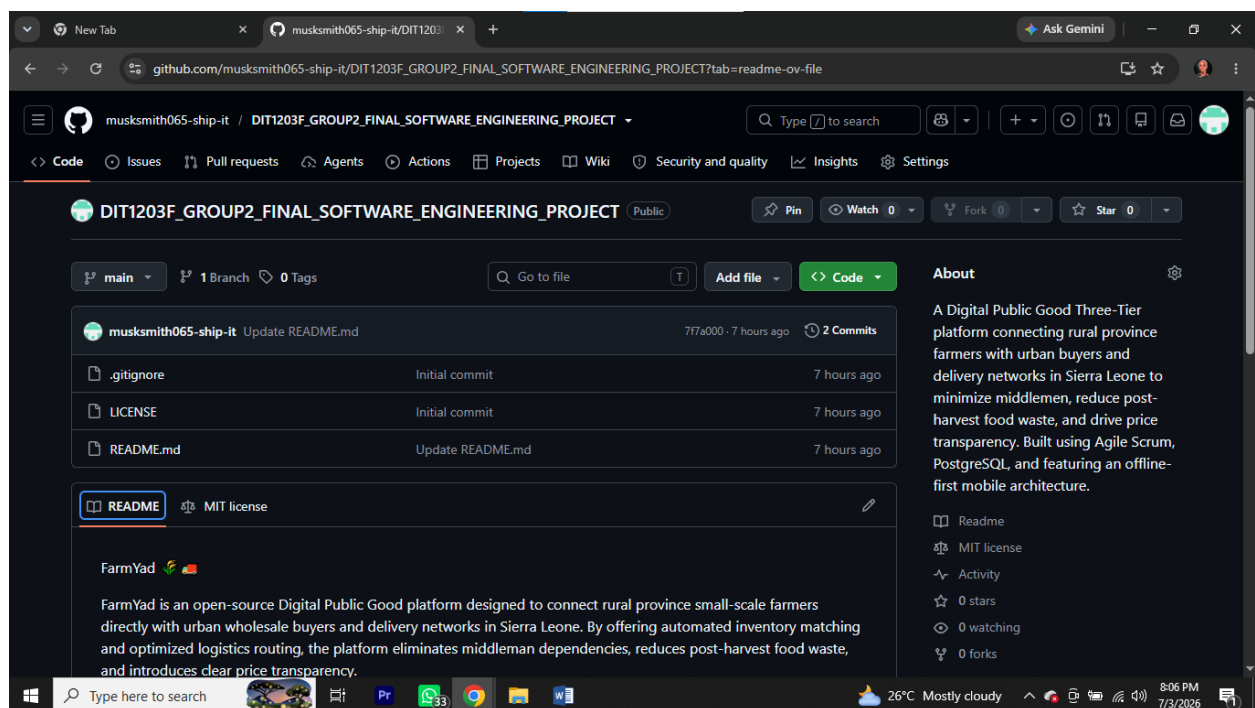
SCREEN 14 (Profile)

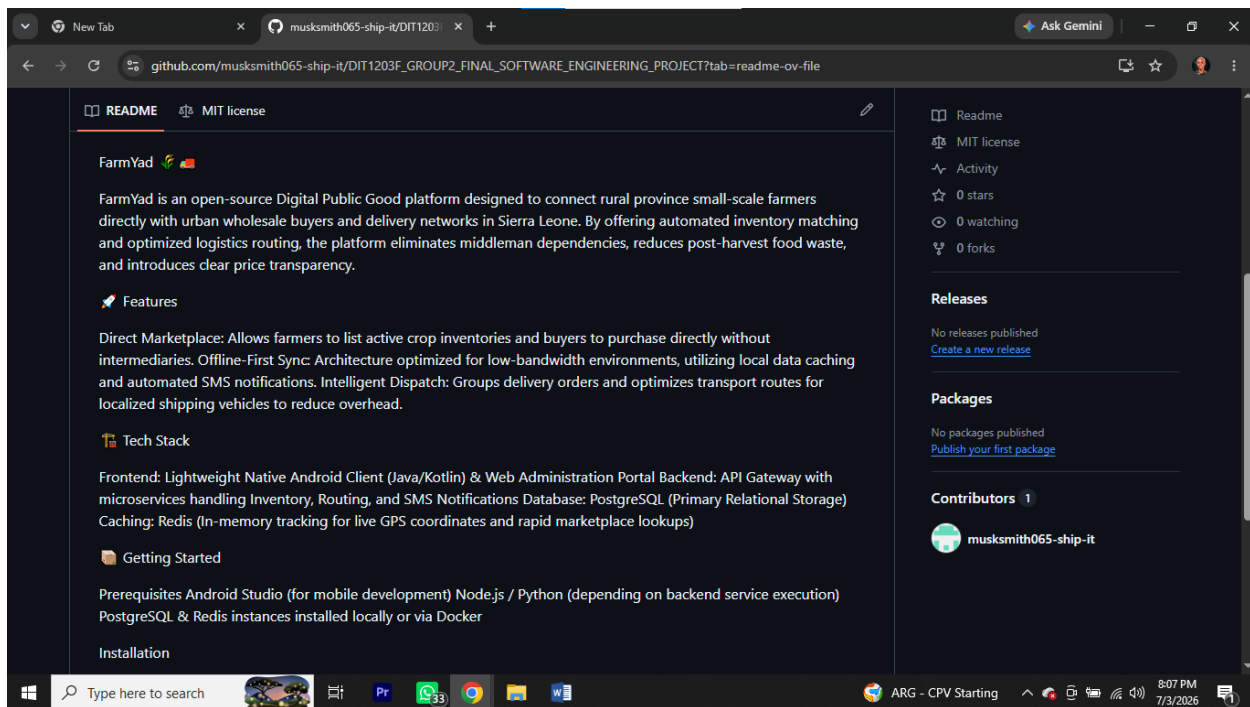


The **Profile screen** is where a user can view and manage their personal or account information within the system.

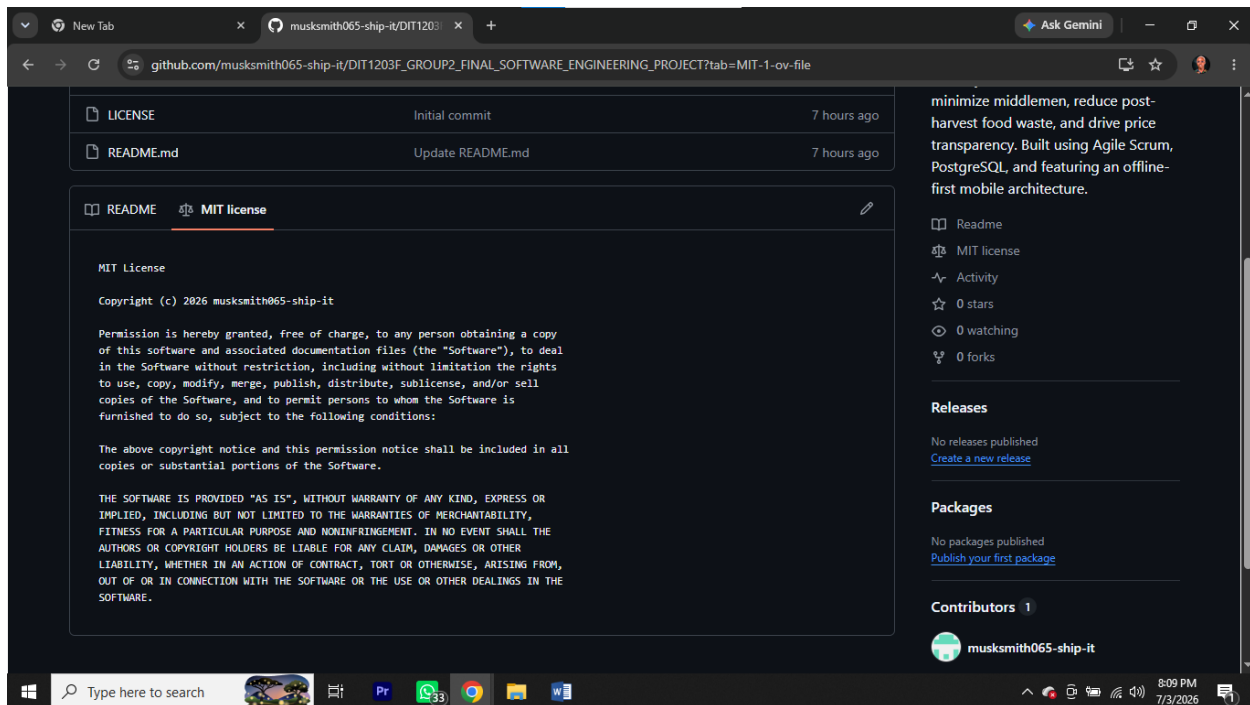
It typically displays details such as the user's name, contact information, role (e.g., farmer or admin), and profile picture. It may also allow updates to settings like password, language, or account preferences.

Overall, it helps users keep their information up to date and personalize their experience in the application.





LICENSE INFORMATION MANAGEMENT



Conclusion

FarmYad is an innovative Digital Public Good platform designed to address the challenges faced by small-scale farmers in Sierra Leone. By connecting farmers directly with buyers through a digital marketplace, the system helps reduce the reliance on middlemen, improves market access, and increases farmers' income. The platform also provides efficient delivery coordination, transparent pricing, and real-time communication between farmers, buyers, and drivers.

The project demonstrates how technology can be used to solve real-world agricultural and economic problems while supporting sustainable development. Through its well-structured architecture, user-friendly interfaces, and integrated logistics services, FarmYad creates a more efficient and reliable agricultural supply chain. Furthermore, the system contributes to the achievement of Sustainable Development Goals (SDGs), particularly SDG 2 (Zero Hunger), SDG 8 (Decent Work and Economic Growth), and SDG 9 (Industry, Innovation, and Infrastructure).

Overall, FarmYad has the potential to transform the agricultural marketplace in Sierra Leone by empowering rural farmers, improving food distribution, and promoting digital innovation in the agricultural sector. The successful implementation of this platform can create lasting social and economic benefits for both producers and consumers.

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